

GP Notification of Death & Final Hospital Summary

To: Dr. R. Patel (Primary Care)

From: Hematology/Oncology Service

Patient: Jason Hernandez, MRN AML079 (DOB 22.11.1971)

Primary diagnosis: Acute Myeloid Leukemia (AML), ELN 2022 Adverse Risk, KMT2A (MLL) rearranged

Initial therapy: Induction "7+3" initiated 15.07.2022

Condolence & Notification

We regret to inform you that **Mr. Jason Hernandez died on 15.08.2023** in our ICU. The immediate cause of death was **septic shock due to multidrug-resistant Pseudomonas aeruginosa bacteremia** in the setting of profound chemotherapy-induced neutropenia. His family was present; code status at the time of death was DNR/DNI per his advance directive.

Diagnostic Background

- **Presentation (July 2022):** Fatigue, gingival bleeding, WBC $46 \times 10^9/L$ with 61% circulating blasts, Hgb 8.2 g/dL, Plt $39 \times 10^9/L$, LDH 784 U/L.
 - **Bone marrow (08.07.2022):** 80–85% myeloblasts; immunophenotype CD13⁺/CD33⁺/CD117⁺/HLA-DR⁺/CD64⁺; aberrant CD7.
 - **Cytogenetics:** 46,XY,t(9;11)(p21;q23) with additional abnormalities on later testing; **KMT2A rearrangement** confirmed by FISH and RNA fusion panel.
 - **Baseline echo:** LVEF 61%. Hepatitis/HIV negative.
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Treatment Course

1. **Induction (15.07.2022):** Cytarabine continuous infusion $\times 7d$ + daunorubicin $\times 3d$ ("7+3").
 - **Complications:** Day 12 neutropenic fever; responsive to piperacillin–tazobactam; mucositis grade 3, TPN $\times 4$ days.
 - **Day 28 marrow (12.08.2022):** Residual disease with 12% blasts (refractory).
 - **Re-induction (FLAG-IDA) 22.08.2022:** Fludarabine + high-dose cytarabine + G-CSF + idarubicin.
 - **Response (late Sept 2022):** Partial cytoreduction; marrow with $\sim 8\%$ blasts.
2. **Bridging/Salvage**
 - We pursued **allogeneic stem cell transplant from 9/10 MUD** after RIC conditioning on 15.09.2022
 - **NGS (Dec 2022):** Persistent KMT2A fusion; no FLT3/NPM1; DNMT3A co-mutation at low VAF emerged.
 - Referred for **clinical trial of a menin inhibitor**; he consented but screen failed (QTc prolongation and rising creatinine).
3. **Documented Relapse:**

- **Bone marrow 19.05.2023:** Re-expansion to 28–32% blasts; escalating transfusion needs; circulating blasts reappeared.
 - Given refractory biology and comorbidities (HTN, class II obesity, mild CKD), we recommended **palliative-intent low-intensity therapy** with infection vigilance and transfusion support while still exploring trial options.
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Final Admission (07.08.2023 → 15.08.2023)

Reason: Fever, rigors, hypotension at home; ANC $0.02 \times 10^9/L$.

Initial labs: WBC 0.5 (ANC 0.02), Hgb 7.1 g/dL, Plt $9 \times 10^9/L$; Cr 1.6 mg/dL; AST/ALT 54/61; lactate 5.1 mmol/L; procalcitonin 12 ng/mL.

Management: Broad-spectrum antibiotics (meropenem → ceftolozane/tazobactam per sensitivities), micafungin, norepinephrine/vasopressin; stress-dose steroids; platelets/RBC transfusions; rasburicase not indicated (urate 3.4).

Microbiology: Central and peripheral blood cultures positive for **P. aeruginosa** (carbapenem-resistant; susceptible to ceftolozane/tazobactam).

Course: Progressive respiratory failure with diffuse bilateral infiltrates (CXR/CT consistent with ARDS vs. infectious pneumonitis), acute kidney injury (Cr peak 2.4). After family meeting, he opted against intubation/CRRT. Comfort-focused measures were instituted. **Pronounced dead at 17:36 on 15.08.2023.**

Medications Near End of Life

- Antimicrobials per sensitivities, G-CSF trialed without count recovery, PRN hydromorphone and lorazepam for dyspnea/anxiety, ondansetron, pantoprazole, stool regimen. Chronic meds (amlodipine) held during shock.
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Comorbidities

- Hypertension (previously on amlodipine 10 mg daily).
 - Obstructive sleep apnea (non-adherent to CPAP).
 - Mild chronic kidney disease (baseline Cr 1.2–1.3).
 - Generalized anxiety disorder.
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Family/Aftercare

Spouse **Elena Hernandez** is next of kin; bereavement resources provided. Autopsy declined. Death certificate completed; copy forwarded to your office.

Please accept our condolences and contact us if you need further documentation.

Sincerely,
[Name], MD
Consultant Hematologist